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Sequential Lighting and Switching Systems

Session 2

SEQUENTIAL LIGHTING AND SWITCHING SYSTEMS

Curiosity Kickoff

- Why do LEDs only glow in one direction (polarity)?
- Why does an LED need a resistor to work safely?
- What happens if you connect an LED directly to a battery without any control?

Concept Connection

This topic introduces students to Programming Basics through the session LED Control. The goal is to keep the learning simple, practical, and beginner-friendly. Students first understand the concept in easy words, then see a small Arduino program, and then connect the code to the real output or input behaviour.

1. Watt SMD LED

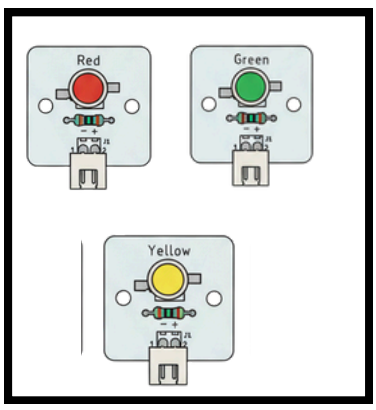
- I'm flat and powerful. Just one of me can light up your whole table like a torch.
- I don't like wasting energy, so I shine super bright without getting too hot.
- You've already met me I live in your mobile flashlight!

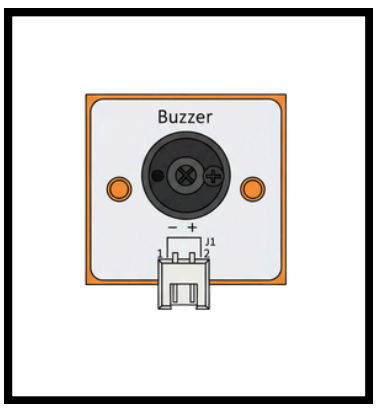
2. Rocker switch PCB

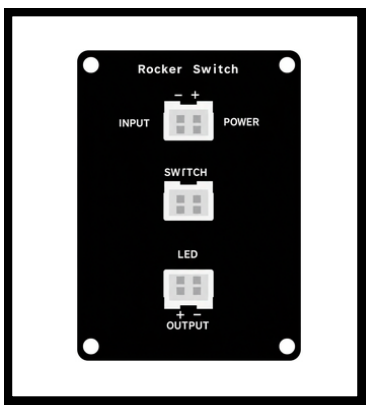
A rocker switch PCB is a small circuit board designed to hold and connect a rocker switch easily in electronic projects.

3. Buzzer

A buzzer is an electronic component that generates sound when electric current flows through it.







Beginner: Blink LED with Delay 3 Patterns

Class Explanation

In this class, students learn the basic idea of Blink LED with Delay 3 Patterns. The code below is kept simple so beginners can focus on understanding the main concept without extra complexity.

Materials Required

- Arduino with attached shield
- 1 × Red 1W SMD LED on breakout board
- RMC wires
- USB cable

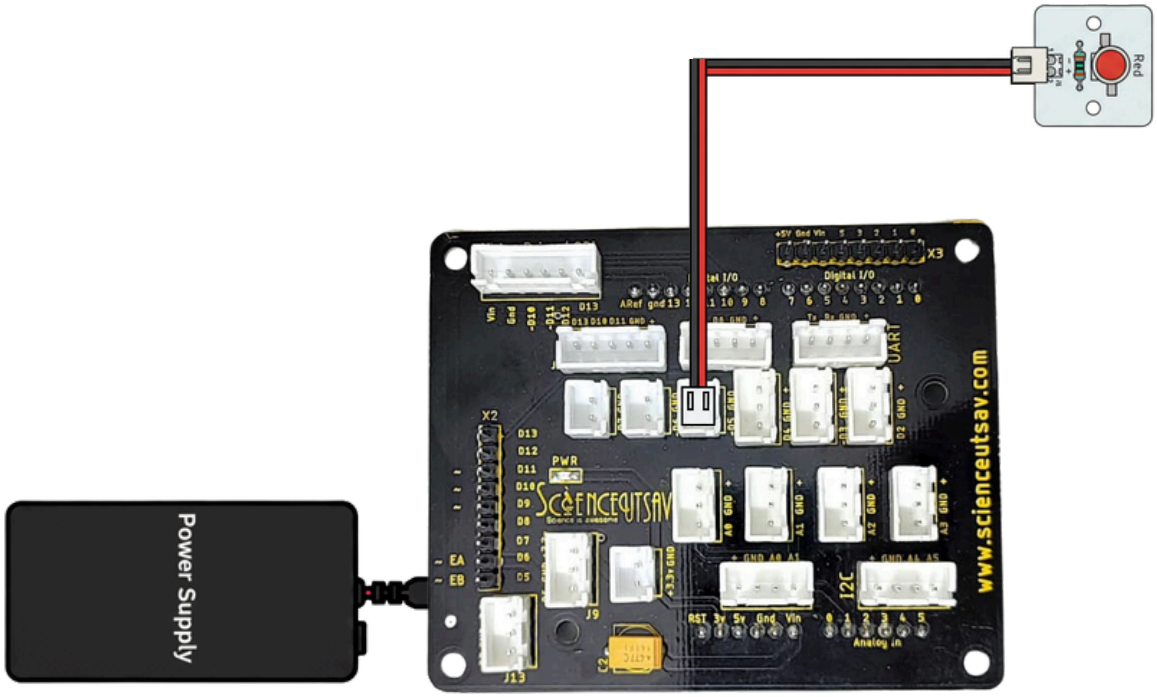
How to Connect

Use RMC wires to connect SMD LED breakout board directly to the Arduino:

- Red SMD LED → Arduino pin D5 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
const int ledPin = 5;

void setup() \
 pinMode(ledPin, OUTPUT);
\

void loop() \
 digitalWrite(ledPin, HIGH);
 delay(200);
 digitalWrite(ledPin, LOW);
 delay(200);
 digitalWrite(ledPin, HIGH);
 delay(500);
 digitalWrite(ledPin, LOW);
 delay(500);
 digitalWrite(ledPin, HIGH);
 delay(1000);
 digitalWrite(ledPin, LOW);
 delay(1000);
\
```

Code Explanation

- In an Arduino program, first we define the pins or variables. This means we give a name to a pin number so it is easy to use in the code. For example, if we write int ledPin = 5;, it means we are using pin number 5 and calling it ledPin.
- Next comes the setup() function. This part runs only one time when the Arduino is turned on. It is used to prepare the pins and start things like the Serial Monitor. Inside setup, we use pinMode() to tell the Arduino what each pin will do. If we write pinMode(ledPin, OUTPUT);, it means the pin will send signals to control devices like an LED.
- After that comes the loop() function. This part runs again and again forever. Arduino keeps repeating whatever is written inside loop, so it can continuously control or check the circuit.
- Inside the loop, we can use digitalWrite() to turn an output ON or OFF. For example, digitalWrite(ledPin, HIGH); will turn the LED ON, and digitalWrite(ledPin, LOW); will turn it OFF.

So overall, the program first sets everything in setup(), then keeps running the loop continuously, controlling devices like LEDs using functions such as pinMode() and digitalWrite().

Try This Yourself Code

This Try This Yourself code keeps the same topic of Blink LED with Delay 3 Patterns but changes the pattern, values, or outputs slightly

Try This Yourself Code

```
const int ledPin = 5;

void setup() \
 pinMode(ledPin, OUTPUT);
\

void loop() \
 digitalWrite(ledPin, HIGH);
 delay(150);
 digitalWrite(ledPin, LOW);
 delay(150);
 digitalWrite(ledPin, HIGH);
 delay(400);
 digitalWrite(ledPin, LOW);
 delay(400);
 digitalWrite(ledPin, HIGH);
 delay(900);
 digitalWrite(ledPin, LOW);
 delay(900);
\
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn

- the basic idea of Blink LED with Delay 3 Patterns how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Intermediate: Blink LED with Delay 4 Patterns

Class Explanation

In this class, students learn the basic idea of Blink LED with Delay 4 Patterns. The code below is kept simple so beginners can focus on understanding the main concept without extra complexity.

Materials Required

- Arduino with attached shield
- 1 × Red 1W SMD LED on breakout board
- 1 × Green 1W SMD LED on breakout board
- RMC wires
- USB cable

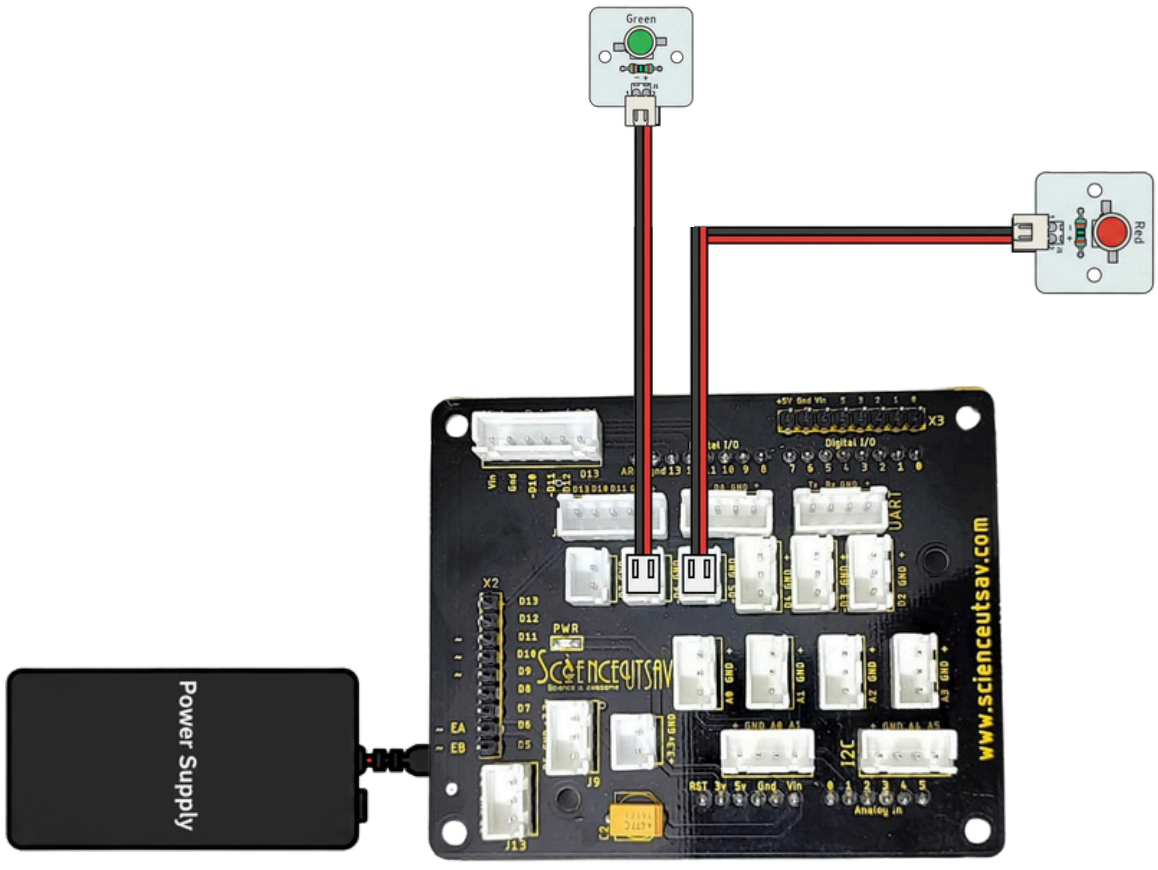
How to Connect

Use RMC wires to connect SMD LED breakout board directly to the Arduino:

- Red SMD LED → Arduino pin D5 and GND
- Green SMD LED → Arduino pin D6 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
const int ledPin = 5;

void setup() \
 pinMode(ledPin, OUTPUT);
\

void loop() \
 digitalWrite(ledPin, HIGH); delay(100); digitalWrite(ledPin, LOW); delay(100);
 digitalWrite(ledPin, HIGH); delay(200); digitalWrite(ledPin, LOW); delay(200);
 digitalWrite(ledPin, HIGH); delay(400); digitalWrite(ledPin, LOW); delay(400);
 digitalWrite(ledPin, HIGH); delay(800); digitalWrite(ledPin, LOW); delay(800);
\
```

Code Explanation

- In an Arduino program, first we define the pins or variables. This means we give a name to a pin number so it is easy to use in the code. For example, if we write int ledPin = 5;, it means we are using pin number 5 and calling it ledPin.

- Next comes the setup() function. This part runs only one time when the Arduino is turned on. It is used to prepare the pins and start things like the Serial Monitor. Inside setup, we use pinMode() to tell the Arduino what each pin will do. If we write pinMode(ledPin, OUTPUT);, it means the pin will send signals to control devices like an LED.

- After that comes the loop() function. This part runs again and again forever. Arduino keeps repeating whatever is written inside loop, so it can continuously control or check the circuit.

- Inside the loop, we can use digitalWrite() to turn an output ON or OFF. For example, digitalWrite(ledPin, HIGH); will turn the LED ON, and digitalWrite(ledPin, LOW); will turn it OFF.

So overall, the program first sets everything in setup(), then keeps running the loop continuously, controlling devices like LEDs using functions such as pinMode() and digitalWrite().

Try This Yourself Code

This Try This Yourself code keeps the same topic of Blink LED with Delay 4 Patterns but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
const int ledPin = 5;

void setup() \
 pinMode(ledPin, OUTPUT);
\

void loop() \
 digitalWrite(ledPin, HIGH);
 delay(100);
 digitalWrite(ledPin, LOW);
 delay(100);
 digitalWrite(ledPin, HIGH);
 delay(300);
 digitalWrite(ledPin, LOW);
 delay(300);
 digitalWrite(ledPin, HIGH);
 delay(600);
 digitalWrite(ledPin, LOW);
 delay(600);
 digitalWrite(ledPin, HIGH);
 delay(1000);
 digitalWrite(ledPin, LOW);
 delay(1000);
\
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn

the basic idea of Blink LED with Delay 4 Patterns how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Advanced: buzzzer system

Class Explanation

In this class, students learn the basic idea of buzzer system. The code below is kept simple so beginners can focus on understanding the main concept without extra complexity.

Materials Required

- Arduino with attached shield
- 1 × Buzzer
- RMC wires
- USB cable

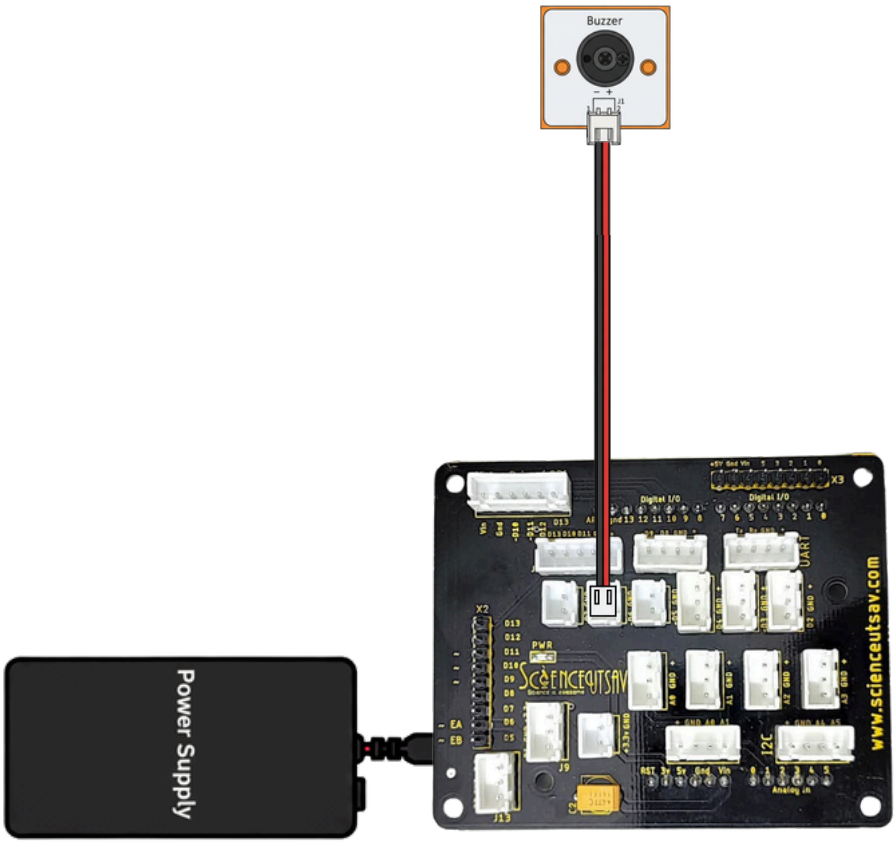
How to Connect

Use RMC wires to connect directly to the Arduino:

- 1 × Buzzer→ Arduino pin D6 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
void setup() {
 pinMode(6, OUTPUT);
}

void loop() {
 digitalWrite(6, HIGH);
 delay(200);

 digitalWrite(6, LOW);
 delay(200);
}
```

Code Explanation

- In this Arduino program, an LED is connected to pin 6.
- The setup() function runs once when the Arduino starts. It sets pin 6 as an OUTPUT using pinMode().
- The loop() function runs continuously. It turns the LED ON using digitalWrite(HIGH) and keeps it ON for 200 milliseconds using delay(200).
- Then, it turns the LED OFF using digitalWrite(LOW) and waits for another 200 milliseconds.
- So overall, the LED continuously blinks by turning ON and OFF every 200 milliseconds.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Logic Based LED Control if else but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
const int buzzer1 = 6;
const int buzzer2 = 7;

void setup() {
  pinMode(buzzer1, OUTPUT);
  pinMode(buzzer2, OUTPUT);
}

void loop() {
  digitalWrite(buzzer1, HIGH);
  digitalWrite(buzzer2, HIGH);
  delay(200);

  digitalWrite(buzzer1, LOW);
  digitalWrite(buzzer2, LOW);
  delay(200);
}
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn

- the basic idea of Logic Based LED Control and buzzer system if else how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Expert: Advanced LED Patterns Loops Arrays

Class Explanation

In this class, students learn the basic idea of Advanced LED Patterns Loops Arrays. The code below is kept simple so beginners can focus on understanding the main concept without extra complexity.

Materials Required

- Arduino with attached shield
- 2 × Red 1W SMD LED on breakout board
- 1 × Green 1W SMD LED on breakout board
- 1 × Rocker switch PCB
- RMC wires
- USB cable

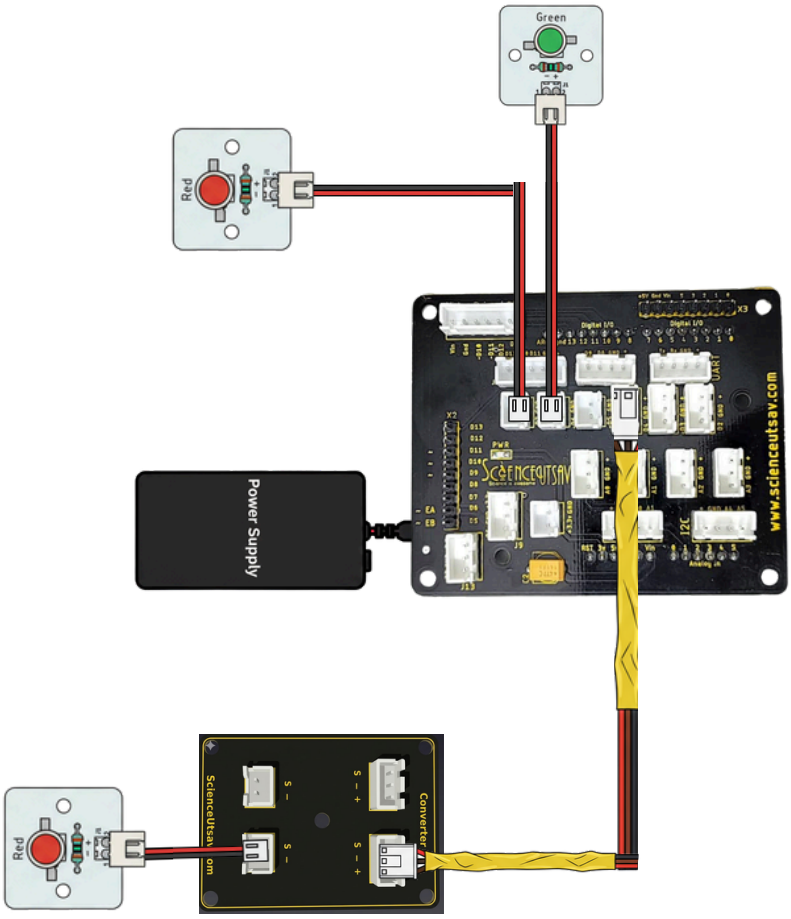
How to Connect

Use RMC wires to connect SMD LED breakout board directly to the Arduino:

- 1 × Red 1W SMD LED → Arduino pin D7 and GND
- 1 × Green 1W SMD LED→ Arduino pin D6 and GND
- 1 × Rocker switch PCB → Arduino pin D5 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
const int led1 = 4;
const int led2 = 6;
const int led3 = 7;

void setup() {
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
  pinMode(led3, OUTPUT);
}

void loop() {
  // All 3 LEDs ON together
  digitalWrite(led1, HIGH);
  digitalWrite(led2, HIGH);
  digitalWrite(led3, HIGH);
  delay(500);

  // OFF one by one
  digitalWrite(led1, LOW);
  delay(300);

  digitalWrite(led2, LOW);
  delay(300);

  digitalWrite(led3, LOW);
  delay(500);
}
```

Code Explanation

- In this Arduino program, three LEDs are connected to pins 4 (led1), 6 (led2), and 7 (led3).
- The setup() function runs once when the Arduino starts. It sets all three LED pins as OUTPUT using pinMode().
- The loop() function runs continuously. First, all three LEDs are turned ON together using digitalWrite(HIGH) and remain ON for 500 milliseconds.
- Then, the LEDs are turned OFF one by one. First, LED1 turns OFF, followed by LED2, and then LED3, with a 300 ms delay between each.
- So overall, all three LEDs light up together and then switch OFF one after another in a sequence.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Advanced LED Patterns Loops Arrays but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
const int led1 = 4;
const int led2 = 5;
const int led3 = 6;
const int led4 = 7;

void setup() {
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
  pinMode(led3, OUTPUT);
  pinMode(led4, OUTPUT);
}

void loop() {
  // All 4 LEDs ON together
  digitalWrite(led1, HIGH);
  digitalWrite(led2, HIGH);
  digitalWrite(led3, HIGH);
  digitalWrite(led4, HIGH);
  delay(500);

  // OFF one by one
  digitalWrite(led1, LOW);
  delay(300);

  digitalWrite(led2, LOW);
  delay(300);

  digitalWrite(led3, LOW);
  delay(300);

  digitalWrite(led4, LOW);
  delay(500);
}
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn

- the basic idea of Advanced LED Patterns Loops Arrays how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Learning Outcomes

After completing this session, students should be able to:

- the basic idea of PWM Basics how the code connects with the circuit behavior how to upload and test a simple Arduino sketch
- the basic idea of LED Fading with PWM how the code connects with the circuit behavior how to upload and test a simple Arduino sketch
- the basic idea of Buzzer Tone Control how the code connects with the circuit behavior how to upload and test a simple Arduino sketch
- the basic idea of Combining LED Dimmer and Buzzer Tones how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Try Out Other Things

- use the Try This Yourself code after each main class code before asking students to build a completely new variation
- ask students to point out the exact lines that changed between the first code and the Try This Yourself codecode
- let students first run the Try This Yourself code code as it is and then change one value, one condition, or one output